

Emad Bahrami

Contact Information

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Research Summary

PhD researcher in computer vision and multimodal AI, working on video understanding, multimodal LLMs, and reinforcement-learning post-training. At Microsoft Research, developed STRIVE, a GRPO-based RL post-training method for video question answering and spatiotemporal reasoning (two US patent applications). Prior work on temporal action segmentation, long-term anticipation, video prediction, and efficient video models, published at CVPR, ICCV, ECCV, AAAI, and IJCV. Currently working on generative world models and visual-physical reasoning.

Education

University of Bonn
PhD in **Computer Science**
Advisor: [Prof. Dr. Jürgen Gall](#)

Bonn, Germany
Jul 2021 – Present

University of Bonn
M.Sc. in **Computer Science**
Thesis advisor: [Prof. Dr. Heiko Röglin](#)

Bonn, Germany
Oct 2015 – Sep 2018

University of Tehran
B.Sc. in **Computer Science**

Tehran, Iran
Sep 2010 – Jul 2015

Professional Experience

- **Research Intern** Nov 2025 – Jan 2026
Toyota Motor Europe, Belgium
 - Developed multi-view video understanding methods for factory tow-car data to improve operational efficiency.
 - Developed a few-shot action segmentation approach for assembly-line videos as the basis for an automated annotation tool to reduce labeling costs; obtained promising preliminary results.
- **Research Intern** May 2025 – Jul 2025
Microsoft Research, Berlin, Germany
 - Developed RL post-training methods (GRPO) for multimodal LLMs, targeting video question answering and spatiotemporal reasoning.
 - Designed [STRIVE](#), an extension of GRPO that incorporates structured spatiotemporal visual variants and visual importance into multimodal RL training.
 - Built multi-GPU training and large-scale video-language evaluation pipelines for multimodal models.
 - Co-inventor on two pending US patent applications on RL post-training of multimodal LLMs.
- **Visiting Researcher** May 2020 – Dec 2020
[Computer Vision Group \(Prof. Gall\)](#), University of Bonn, Germany
 - Developed adaptive temporal feature-resolution mechanisms for 3D CNNs that dynamically allocate computation across video segments, reducing computational cost while maintaining recognition performance (CVPR 2021).

- **Research Associate** Jan 2019 – Jun 2021
[Deep MI Lab](#), German Center for Neurodegenerative Diseases (DZNE), Bonn, Germany
 - Co-developed deep-learning pipelines for fine-grained brain MRI segmentation: CerebNet (cerebellum) and FastSurfer-HypVINN (hypothalamus), released as part of the open-source FastSurfer toolkit (*NeuroImage, Imaging Neuroscience*).
- **Research Assistant** Jan 2016 – Oct 2018
[Max Planck Institute for Research on Collective Goods](#), Bonn, Germany

Publications

- **STRIVE: Structured Spatiotemporal Exploration for Reinforcement Learning in Video Question Answering**
 Bahrami E.[†], Zatsarynna O., Pathak P., Sengupta S., Gall J., Fayyaz M., *arXiv 2026, under review* [[pdf](#)]
[†] Work done during an internship at Microsoft Research.
- **Post-training Video Language Models for Mistake Recognition**
 Spurio F., Zatsarynna O., Doorenbos L., Bahrami E., Francesca G., Gall J., *BMVC 2026* [[pdf](#)]
- **Sequence-Adaptive Video Prediction in Continuous Streams using Diffusion Noise Optimization**
 Mokhtarzadeh Azar S., Bahrami E., Pallotta E., Francesca G., Timofte R., Gall J., *CVIU 2026* [[pdf](#)]
- **Looking into the Unknown: Exploring Action Discovery for Segmentation of Known and Unknown Actions**
 Spurio F., Bahrami E., Zatsarynna O., Abu Farha Y., Francesca G., Gall J., *CVIU 2026* [[pdf](#)]
- **Towards Generalizing Temporal Action Segmentation to Unseen Views**
 Bahrami E.*, Zatsarynna O.*, Francesca G., Gall J., *IJCV 2025* [[pdf](#)]
 * Equal contribution.
- **MANTA: Diffusion Mamba for Efficient and Effective Stochastic Long-Term Dense Anticipation**
 Zatsarynna O., Bahrami E., Abu Farha Y., Francesca G., Gall J., *CVPR 2025* [[pdf](#)]
- **Hierarchical Vector Quantization for Unsupervised Action Segmentation**
 Spurio F., Bahrami E., Francesca G., Gall J., *AAAI 2025* [[pdf](#)]
- **Gated Temporal Diffusion for Stochastic Long-Term Dense Anticipation**
 Zatsarynna O.*, Bahrami E.*, Abu Farha Y., Francesca G., Gall J., *ECCV 2024* [[pdf](#)]
 * Equal contribution.
- **How Much Temporal Long-Term Context is Needed for Action Segmentation?**
 Bahrami E., Francesca G., Gall J., *ICCV 2023* [[pdf](#)]
- **FastSurfer-HypVINN: Automated Sub-segmentation of the Hypothalamus and Adjacent Structures on High-resolutional Brain MRI**
 Estrada S., K gler D., Bahrami E., Mousa D., Xu P., Breteler M., Aziz A., Reuter M., *Imaging Neuroscience 2023* [[pdf](#)]
- **Robust Action Segmentation from Timestamp Supervision**
 Souri Y.*, Abu Farha Y.*, Bahrami E.*, Francesca G., Gall J., *BMVC 2022* [[pdf](#)]
 * Equal contribution.

- **TaylorSwiftNet: Taylor Driven Temporal Modeling for Swift Future Frame Prediction**
 Pourheydari S.*, **Bahrami E.***, Fayyaz M.*, Francesca G., Noroozi M., Gall J., *BMVC 2022* [[pdf](#)]
 * Equal contribution.
- **CerebNet: A Fast and Reliable Deep-learning Pipeline for Detailed Cerebellum Sub-segmentation**
 Faber J.*, Kügler D.*, **Bahrami E.***, *et al.*, *NeuroImage 2022* [[pdf](#)]
 * Equal contribution.
- **3D CNNs with Adaptive Temporal Feature Resolutions**
 Fayyaz M.*, **Bahrami E.***, Diba A., Noroozi M., Adeli E., Van Gool L., Gall J., *CVPR 2021* [[pdf](#)]
 * Equal contribution.

Patent Applications

Two US patent applications (pending) on reinforcement-learning post-training of multimodal LLMs, filed by Microsoft.
 Co-inventors: M. Fayyaz, **E. Bahrami**, P. Pathak, S. Sengupta
 Method and System for Predicting Patterns in Future Frames.
 G. Francesca, **E. Bahrami**, O. Zatsarynna, J. Gall
European Patent Office (EPO), application no. EP24187682.0
 Methods for Action Segmentation of a Video Sequence.
 G. Francesca, L. Garattoni, **E. Bahrami**, J. Gall
European Patent Office (EPO), application no. EP23181135.7

Technical Skills

- **Machine Learning:** PyTorch, Hugging Face Transformers, multimodal LLMs/VLMs, RL post-training (GRPO), diffusion and generative models
- **Training & Systems:** distributed multi-GPU training, DeepSpeed, mixed precision, CUDA, Slurm, Linux
- **Efficient ML:** adaptive inference, temporal sparsification, token/feature selection
- **Programming:** Python, Bash, Git

Service & Awards

- [Outstanding Reviewer, CVPR 2025](#)
- Reviewer: CVPR, ICCV, NeurIPS, ICML, ICLR, TPAMI